

Group Messenger

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Often, we communicate within a known group of office or friends using short key words or phrases. This group messenger can be useful to such small groups for whom a keyword is sufficient to understand the message. The display shows current date and time by default when there is no message and serves as a decorative piece on worktable.

This messenger can hold up to ten messages. On receipt of eleventh message, the oldest message is deleted and the latest is added to the list of messages, using the first-in-first-out principle. It may be used by up to eight persons to communicate with each other using keywords.

Circuit and working

Circuit diagram of the group messenger is shown in Fig. 1. It comprises a 40-pin AVR microcontroller (ATmega32), a long-range wireless transceiver (HC-12), an RTC IC (DS1307), and a 5V regulator IC (7805). A 9V or 12V DC adaptor, or a battery, may be used as power supply for the circuit. All the switches ES1 through ES5 and SS1 through SS8 are tied to a 470-ohm resistor (R28) to avoid over current flow through any pin of the microcontroller port.

ATmega32A has the required features for this project. It has 32kB flash program memory to store the hex file,

1kB EEPROM to store usernames, 2kB internal SRAM to run the program and hold the values of variables/buffers, two-wire serial interface for RTC IC DS1307 interface, programmable serial USART for HC-12 wireless transceiver, and 32 programmable I/O lines for various switches and other outputs/indicators.

HC-12 is a 5-pin, small-size (28mm × 15mm × 4mm approx.) wireless transceiver with antenna. It uses USART port for communicating with the microcontroller (MCU). It has 433.4MHz to 473.0 MHz carrier band with multiple channels to set, with a total of 100 channels with channel stepping of 400kHz, and works on

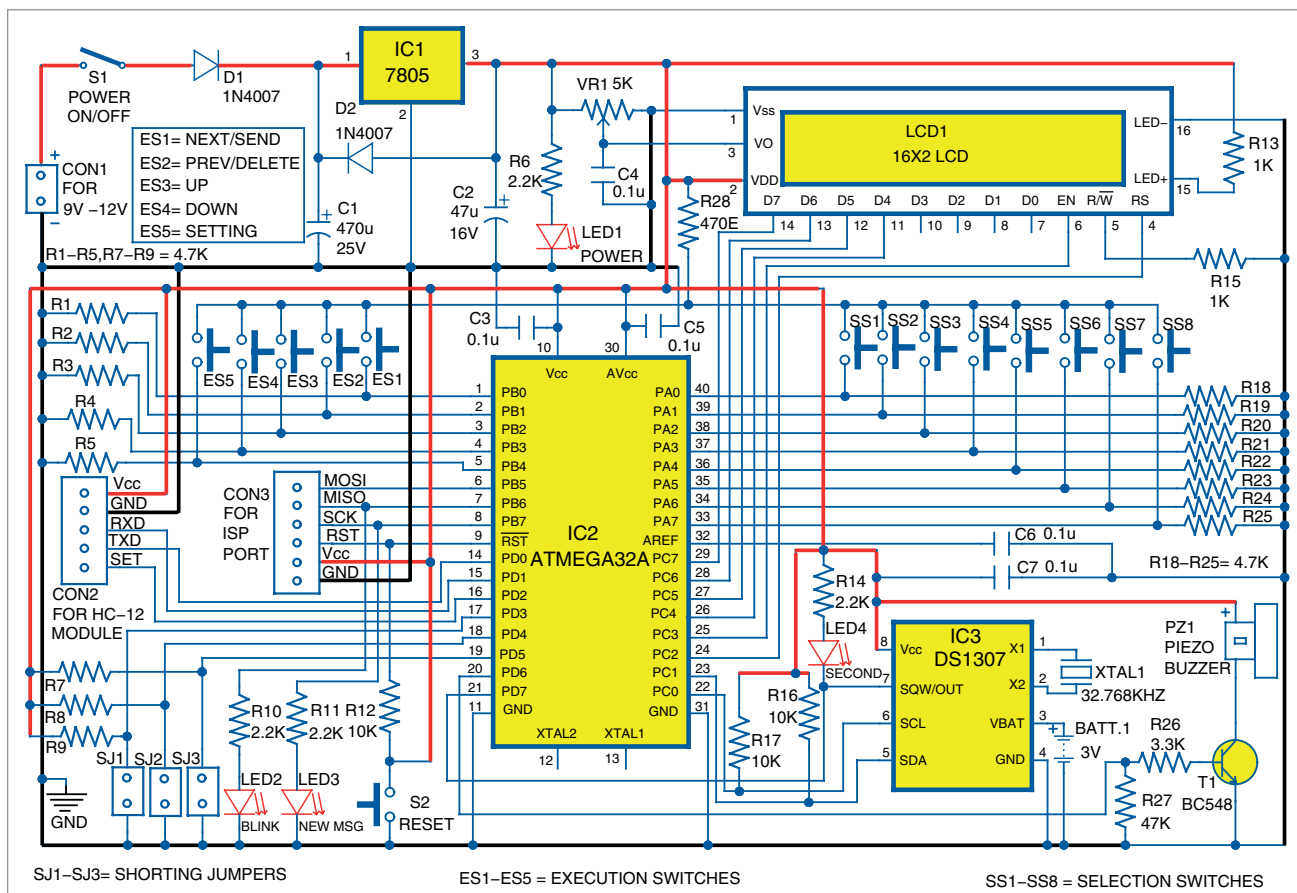


Fig. 1: Circuit diagram of group messenger